Problem 10: Subscription & Auto-Payment Tracker – Mumbai, Maharashtra

1. Introduction

- India's digital subscription economy is growing rapidly
  - OTT, music, cloud, fitness apps, SaaS tools — all charge automatically
- Auto-payment (UPI/credit card) makes renewals seamless but invisible
- Users lose track of active subscriptions over time
- No centralized system exists to monitor recurring digital payments
- Banking apps only show raw transactions — no recurrence detection

2. Problem Statement

- No intelligent system to track, alert, and manage recurring subscriptions
- Key gaps:
  - Cannot detect recurring payments from raw transaction data
  - No alerts before auto-deductions happen
  - No visibility into unused or forgotten subscriptions
  - No monthly subscription expense summary
  - No optimization or cancellation suggestions

3. Real-World Impact

- Users unknowingly pay for services they stopped using
- Trial plans silently convert to paid — user unaware
- Overlapping family subscriptions go undetected
- Money leaks silently every month — noticed only in bank statements
- Causes financial stress and loss of trust in digital payments

4. Example Scenario (Expanded)

- User: Neha, working professional, Mumbai
- Subscriptions: Netflix, Spotify, iCloud, Hotstar, Gym app, 2 SaaS tools
- All set on auto-payment via UPI / credit card
- She cancels gym physically but forgets to cancel the app membership
- Stops using one SaaS tool — subscription still active and charging

- ₹2,000–₹3,000 silently deducted every month
- Notices only after 3 months — while reviewing bank statement
- No alert sent, no system flagged inactivity, no cancellation prompt

5. Detailed Problems Identified

- No pre-deduction alerts or renewal reminders
- Cannot detect recurring patterns from bank transactions
- No subscription categorization (entertainment / productivity / health)
- No tracking of billing cycles or renewal dates
- Cannot identify dormant / unused subscriptions
- No monthly budget summary for subscription expenses
- No cancellation guidance or cost-saving suggestions

6. Existing System Limitations

- Banking apps — show raw transactions, no recurrence detection
- UPI apps — do not track subscription lifecycles or renewal dates
- Expense tracker apps — require 100% manual entry, no auto-detection
- No cross-platform aggregation across UPI, cards, wallets
- No intelligent inference of subscription status from payment data

7. Challenges

- Detecting recurring payments from inconsistent transaction descriptions
- Handling varied billing cycles — monthly, quarterly, annual
- Distinguishing one-time payments from subscriptions
- Defining 'inactivity' without direct access to app usage data
- Sending timely alerts without overwhelming users with notifications
- Supporting multiple payment modes — UPI, credit card, wallet

8. Constraints & Assumptions

- Hackathon scope — full deployment not required
- No paid APIs or licensed tools used
- Mock transaction datasets used as input

- System design and logic prioritized over complete UI
- Assumes structured transaction data is available
- Recurrence detection based on amount consistency and frequency patterns

9. Why This Problem Matters

- India's digital subscription market growing exponentially
- Financial leakage from unused subscriptions impacts millions
- A smart tracker builds financial awareness and spending discipline
- Fintech products gain user trust by being proactive, not reactive
- Directly addresses a real, daily pain point for urban digital users

Conclusion

- A Subscription & Auto-Payment Tracker solves a silent but costly problem
- Key capabilities needed:
  - Auto-detect recurring payments
  - Alert users before renewal deductions
  - Identify and flag unused subscriptions
  - Show monthly spend dashboard
  - Suggest cancellations and optimizations
- Outcome: Users gain full control over recurring digital expenses
- Impact: Prevents financial drain, builds smarter spending habits